

Kevin_Hooman_Assignment3.1

May 25, 2026

```
[27]: # Homework 3 _ Feature Group
```

```
[3]: #!pip install --force-reinstall 'sagemaker>=2.200,<3'
```

```
Collecting sagemaker<3,>=2.200
  Downloading sagemaker-2.257.3-py3-none-any.whl.metadata (17 kB)
Collecting attrs<26,>=24 (from sagemaker<3,>=2.200)
  Downloading attrs-25.4.0-py3-none-any.whl.metadata (10 kB)
Collecting boto3<2.0,>=1.42.2 (from sagemaker<3,>=2.200)
  Using cached boto3-1.43.14-py3-none-any.whl.metadata (6.6 kB)
Collecting cloudpickle>=2.2.1 (from sagemaker<3,>=2.200)
  Using cached cloudpickle-3.1.2-py3-none-any.whl.metadata (7.1 kB)
Collecting docker (from sagemaker<3,>=2.200)
  Using cached docker-7.1.0-py3-none-any.whl.metadata (3.8 kB)
Collecting fastapi (from sagemaker<3,>=2.200)
  Using cached fastapi-0.136.3-py3-none-any.whl.metadata (27 kB)
Collecting google-pasta (from sagemaker<3,>=2.200)
  Downloading google_pasta-0.2.0-py3-none-any.whl.metadata (814 bytes)
Collecting graphene<4,>=3 (from sagemaker<3,>=2.200)
  Using cached graphene-3.4.3-py2.py3-none-any.whl.metadata (6.9 kB)
Collecting importlib-metadata<7.0,>=1.4.0 (from sagemaker<3,>=2.200)
  Using cached importlib_metadata-6.11.0-py3-none-any.whl.metadata (4.9 kB)
Collecting jsonschema (from sagemaker<3,>=2.200)
  Using cached jsonschema-4.26.0-py3-none-any.whl.metadata (7.6 kB)
Collecting numpy<3.0,>=1.26.4 (from sagemaker<3,>=2.200)
  Using cached
numpy-2.4.6-cp312-cp312-manylinux_2_27_x86_64.manylinux_2_28_x86_64.whl.metadata
(6.6 kB)
Collecting omegaconf<3,>=2.2 (from sagemaker<3,>=2.200)
  Using cached omegaconf-2.3.0-py3-none-any.whl.metadata (3.9 kB)
Collecting packaging<25,>=23.0 (from sagemaker<3,>=2.200)
  Downloading packaging-24.2-py3-none-any.whl.metadata (3.2 kB)
```

Collecting pandas>=2.3.0 (from sagemaker<3,>=2.200)
 Using cached pandas-3.0.3-cp312-cp312-manylinux_2_24_x86_64.manylinux_2_28_x86_64.whl.metadata (79 kB)

Collecting pathos (from sagemaker<3,>=2.200)
 Downloading pathos-0.3.5-py3-none-any.whl.metadata (11 kB)

Collecting platformdirs (from sagemaker<3,>=2.200)
 Using cached platformdirs-4.9.6-py3-none-any.whl.metadata (4.7 kB)

Collecting protobuf<7.0,>=3.12 (from sagemaker<3,>=2.200)
 Using cached protobuf-6.33.6-cp39-abi3-manylinux2014_x86_64.whl.metadata (593 bytes)

Collecting psutil (from sagemaker<3,>=2.200)
 Using cached psutil-7.2.2-cp36-abi3-manylinux2010_x86_64.manylinux_2_12_x86_64.manylinux_2_28_x86_64.whl.metadata (22 kB)

Collecting pyyaml>=6.0.1 (from sagemaker<3,>=2.200)
 Using cached pyyaml-6.0.3-cp312-cp312-manylinux2014_x86_64.manylinux_2_17_x86_64.manylinux_2_28_x86_64.whl.metadata (2.4 kB)

Collecting requests (from sagemaker<3,>=2.200)
 Using cached requests-2.34.2-py3-none-any.whl.metadata (4.8 kB)

Collecting sagemaker-core<2.0.0,>=1.0.71 (from sagemaker<3,>=2.200)
 Downloading sagemaker_core-1.0.78-py3-none-any.whl.metadata (4.9 kB)

Collecting schema (from sagemaker<3,>=2.200)
 Using cached schema-0.7.8-py2.py3-none-any.whl.metadata (34 kB)

Collecting smdebug-rulesconfig==1.0.1 (from sagemaker<3,>=2.200)
 Using cached smdebug_rulesconfig-1.0.1-py2.py3-none-any.whl.metadata (943 bytes)

Collecting tblib<4,>=1.7.0 (from sagemaker<3,>=2.200)
 Using cached tblib-3.2.2-py3-none-any.whl.metadata (27 kB)

Collecting tqdm (from sagemaker<3,>=2.200)
 Using cached tqdm-4.67.3-py3-none-any.whl.metadata (57 kB)

Collecting urllib3<3.0.0,>=1.26.8 (from sagemaker<3,>=2.200)
 Using cached urllib3-2.7.0-py3-none-any.whl.metadata (6.9 kB)

Collecting uvicorn (from sagemaker<3,>=2.200)
 Using cached uvicorn-0.48.0-py3-none-any.whl.metadata (6.7 kB)

Collecting botocore<1.44.0,>=1.43.14 (from boto3<2.0,>=1.42.2->sagemaker<3,>=2.200)
 Using cached botocore-1.43.14-py3-none-any.whl.metadata (5.6 kB)

Collecting jmespath<2.0.0,>=0.7.1 (from boto3<2.0,>=1.42.2->sagemaker<3,>=2.200)
 Using cached jmespath-1.1.0-py3-none-any.whl.metadata (7.6 kB)

Collecting s3transfer<0.18.0,>=0.17.0 (from boto3<2.0,>=1.42.2->sagemaker<3,>=2.200)
 Using cached s3transfer-0.17.0-py3-none-any.whl.metadata (1.7 kB)

Collecting python-dateutil<3.0.0,>=2.1 (from
 botocore<1.44.0,>=1.43.14->boto3<2.0,>=1.42.2->sagemaker<3,>=2.200)
 Using cached python_dateutil-2.9.0.post0-py2.py3-none-any.whl.metadata (8.4
 kB)

Collecting graphql-core<3.3,>=3.1 (from graphene<4,>=3->sagemaker<3,>=2.200)
 Using cached graphql_core-3.2.8-py3-none-any.whl.metadata (11 kB)
 Collecting graphql-relay<3.3,>=3.1 (from graphene<4,>=3->sagemaker<3,>=2.200)
 Using cached graphql_relay-3.2.0-py3-none-any.whl.metadata (12 kB)
 Collecting typing-extensions<5,>=4.7.1 (from
 graphene<4,>=3->sagemaker<3,>=2.200)
 Using cached typing_extensions-4.15.0-py3-none-any.whl.metadata (3.3 kB)

Collecting zipp>=0.5 (from importlib-metadata<7.0,>=1.4.0->sagemaker<3,>=2.200)
 Using cached zipp-4.1.0-py3-none-any.whl.metadata (3.6 kB)
 Collecting antlr4-python3-runtime==4.9.* (from
 omegaconf<3,>=2.2->sagemaker<3,>=2.200)
 Using cached antlr4_python3_runtime-4.9.3-py3-none-any.whl

Collecting six>=1.5 (from python-dateutil<3.0.0,>=2.1-
 >botocore<1.44.0,>=1.43.14->boto3<2.0,>=1.42.2->sagemaker<3,>=2.200)
 Using cached six-1.17.0-py2.py3-none-any.whl.metadata (1.7 kB)

Collecting pydantic<3.0.0,>=2.0.0 (from sagemaker-
 core<2.0.0,>=1.0.71->sagemaker<3,>=2.200)
 Using cached pydantic-2.13.4-py3-none-any.whl.metadata (109 kB)
 Collecting rich<15.0.0,>=13.0.0 (from sagemaker-
 core<2.0.0,>=1.0.71->sagemaker<3,>=2.200)
 Using cached rich-14.3.4-py3-none-any.whl.metadata (18 kB)
 Collecting mock<5.0,>4.0 (from sagemaker-
 core<2.0.0,>=1.0.71->sagemaker<3,>=2.200)
 Using cached mock-4.0.3-py3-none-any.whl.metadata (2.8 kB)
 Collecting jsonschema-specifications>=2023.03.6 (from
 jsonschema->sagemaker<3,>=2.200)
 Using cached jsonschema_specifications-2025.9.1-py3-none-any.whl.metadata (2.9
 kB)

Collecting referencing>=0.28.4 (from jsonschema->sagemaker<3,>=2.200)
 Using cached referencing-0.37.0-py3-none-any.whl.metadata (2.8 kB)

Collecting rpds-py>=0.25.0 (from jsonschema->sagemaker<3,>=2.200)
 Using cached rpds_py-0.30.0-cp312-cp312-
 manylinux_2_17_x86_64.manylinux2014_x86_64.whl.metadata (4.1 kB)
 Collecting annotated-types>=0.6.0 (from pydantic<3.0.0,>=2.0.0->sagemaker-
 core<2.0.0,>=1.0.71->sagemaker<3,>=2.200)
 Using cached annotated_types-0.7.0-py3-none-any.whl.metadata (15 kB)

Collecting pydantic-core==2.46.4 (from pydantic<3.0.0,>=2.0.0->sagemaker-
 core<2.0.0,>=1.0.71->sagemaker<3,>=2.200)
 Using cached pydantic_core-2.46.4-cp312-cp312-
 manylinux_2_17_x86_64.manylinux2014_x86_64.whl.metadata (6.6 kB)

```

Collecting typing-inspection>=0.4.2 (from pydantic<3.0.0,>=2.0.0->sagemaker-
core<2.0.0,>=1.0.71->sagemaker<3,>=2.200)
  Using cached typing_inspection-0.4.2-py3-none-any.whl.metadata (2.6 kB)
Collecting markdown-it-py>=2.2.0 (from rich<15.0.0,>=13.0.0->sagemaker-
core<2.0.0,>=1.0.71->sagemaker<3,>=2.200)
  Using cached markdown_it_py-4.2.0-py3-none-any.whl.metadata (7.4 kB)
Collecting pygments<3.0.0,>=2.13.0 (from rich<15.0.0,>=13.0.0->sagemaker-
core<2.0.0,>=1.0.71->sagemaker<3,>=2.200)
  Using cached pygments-2.20.0-py3-none-any.whl.metadata (2.5 kB)
Collecting mdurl~=0.1 (from markdown-it-
py>=2.2.0->rich<15.0.0,>=13.0.0->sagemaker-
core<2.0.0,>=1.0.71->sagemaker<3,>=2.200)
  Using cached mdurl-0.1.2-py3-none-any.whl.metadata (1.6 kB)
Collecting charset_normalizer<4,>=2 (from requests->sagemaker<3,>=2.200)
  Using cached charset_normalizer-3.4.7-cp312-cp312-
manylinux2014_x86_64.manylinux_2_17_x86_64.manylinux_2_28_x86_64.whl.metadata
(40 kB)
Collecting idna<4,>=2.5 (from requests->sagemaker<3,>=2.200)
  Using cached idna-3.16-py3-none-any.whl.metadata (6.4 kB)
Collecting certifi>=2023.5.7 (from requests->sagemaker<3,>=2.200)
  Using cached certifi-2026.5.20-py3-none-any.whl.metadata (2.5 kB)
Collecting starlette>=0.46.0 (from fastapi->sagemaker<3,>=2.200)
  Downloading starlette-1.1.0-py3-none-any.whl.metadata (6.3 kB)
Collecting annotated-doc>=0.0.2 (from fastapi->sagemaker<3,>=2.200)
  Using cached annotated_doc-0.0.4-py3-none-any.whl.metadata (6.6 kB)
Collecting anyio<5,>=3.6.2 (from
starlette>=0.46.0->fastapi->sagemaker<3,>=2.200)
  Using cached anyio-4.13.0-py3-none-any.whl.metadata (4.5 kB)
Collecting ppft>=1.7.8 (from pathos->sagemaker<3,>=2.200)
  Downloading ppft-1.7.8-py3-none-any.whl.metadata (12 kB)
Collecting dill>=0.4.1 (from pathos->sagemaker<3,>=2.200)
  Downloading dill-0.4.1-py3-none-any.whl.metadata (10 kB)
Collecting pox>=0.3.7 (from pathos->sagemaker<3,>=2.200)
  Downloading pox-0.3.7-py3-none-any.whl.metadata (8.0 kB)
Collecting multiprocessing>=0.70.19 (from pathos->sagemaker<3,>=2.200)
  Downloading multiprocessing-0.70.19-py312-none-any.whl.metadata (7.5 kB)
Collecting click>=7.0 (from uvicorn->sagemaker<3,>=2.200)
  Using cached click-8.4.1-py3-none-any.whl.metadata (2.6 kB)
Collecting h11>=0.8 (from uvicorn->sagemaker<3,>=2.200)
  Using cached h11-0.16.0-py3-none-any.whl.metadata (8.3 kB)
Downloading sagemaker-2.257.3-py3-none-any.whl (1.7 MB)

```

0.0/1.7 MB

? eta -:--:--

1.7/1.7 MB

106.3 MB/s 0:00:00

Using cached smdebug_rulesconfig-1.0.1-py2.py3-none-any.whl (20 kB)

Downloading attrrs-25.4.0-py3-none-any.whl (67 kB)

Using cached boto3-1.43.14-py3-none-any.whl (140 kB)

Using cached botocore-1.43.14-py3-none-any.whl (15.1 MB)

Using cached graphene-3.4.3-py2.py3-none-any.whl (114 kB)

Using cached graphql_core-3.2.8-py3-none-any.whl (207 kB)

Using cached graphql_relay-3.2.0-py3-none-any.whl (16 kB)

Using cached importlib_metadata-6.11.0-py3-none-any.whl (23 kB)

Using cached jmespath-1.1.0-py3-none-any.whl (20 kB)

Using cached

numpy-2.4.6-cp312-cp312-manylinux_2_27_x86_64.manylinux_2_28_x86_64.whl (16.6 MB)

Using cached omegaconf-2.3.0-py3-none-any.whl (79 kB)

Downloading packaging-24.2-py3-none-any.whl (65 kB)

Using cached protobuf-6.33.6-cp39-abi3-manylinux2014_x86_64.whl (323 kB)

Using cached python_dateutil-2.9.0.post0-py2.py3-none-any.whl (229 kB)

Using cached s3transfer-0.17.0-py3-none-any.whl (86 kB)

Downloading sagemaker_core-1.0.78-py3-none-any.whl (444 kB)

Using cached jsonschema-4.26.0-py3-none-any.whl (90 kB)

Using cached mock-4.0.3-py3-none-any.whl (28 kB)

Using cached platformdirs-4.9.6-py3-none-any.whl (21 kB)

Using cached pydantic-2.13.4-py3-none-any.whl (472 kB)

Using cached

pydantic_core-2.46.4-cp312-cp312-manylinux_2_17_x86_64.manylinux2014_x86_64.whl (2.1 MB)

Using cached pyyaml-6.0.3-cp312-cp312-

manylinux2014_x86_64.manylinux_2_17_x86_64.manylinux_2_28_x86_64.whl (807 kB)

Using cached rich-14.3.4-py3-none-any.whl (310 kB)

Using cached pygments-2.20.0-py3-none-any.whl (1.2 MB)

Using cached tblib-3.2.2-py3-none-any.whl (12 kB)

Using cached typing_extensions-4.15.0-py3-none-any.whl (44 kB)

Using cached urllib3-2.7.0-py3-none-any.whl (131 kB)

Using cached annotated_types-0.7.0-py3-none-any.whl (13 kB)

Using cached cloudpickle-3.1.2-py3-none-any.whl (22 kB)

Using cached jsonschema_specifications-2025.9.1-py3-none-any.whl (18 kB)

Using cached markdown_it_py-4.2.0-py3-none-any.whl (91 kB)

Using cached mdurl-0.1.2-py3-none-any.whl (10.0 kB)

Downloading

pandas-3.0.3-cp312-cp312-manylinux_2_24_x86_64.manylinux_2_28_x86_64.whl (10.9 MB)

0.0/10.9 MB

? eta -:--:--

10.9/10.9 MB

133.1 MB/s 0:00:00

Using cached referencing-0.37.0-py3-none-any.whl (26 kB)
Using cached
rpds_py-0.30.0-cp312-cp312-manylinux_2_17_x86_64.manylinux2014_x86_64.whl (394 kB)
Using cached six-1.17.0-py2.py3-none-any.whl (11 kB)
Using cached typing_inspection-0.4.2-py3-none-any.whl (14 kB)
Using cached zipp-4.1.0-py3-none-any.whl (10 kB)
Using cached docker-7.1.0-py3-none-any.whl (147 kB)
Using cached requests-2.34.2-py3-none-any.whl (73 kB)
Using cached charset_normalizer-3.4.7-cp312-cp312-manylinux2014_x86_64.manylinux_2_17_x86_64.manylinux_2_28_x86_64.whl (216 kB)
Using cached idna-3.16-py3-none-any.whl (74 kB)
Using cached certifi-2026.5.20-py3-none-any.whl (134 kB)
Using cached fastapi-0.136.3-py3-none-any.whl (117 kB)
Using cached annotated_doc-0.0.4-py3-none-any.whl (5.3 kB)
Downloading starlette-1.1.0-py3-none-any.whl (72 kB)
Using cached anyio-4.13.0-py3-none-any.whl (114 kB)
Downloading google_pasta-0.2.0-py3-none-any.whl (57 kB)

Downloading pathos-0.3.5-py3-none-any.whl (82 kB)
Downloading dill-0.4.1-py3-none-any.whl (120 kB)
Downloading multiprocessing-0.70.19-py312-none-any.whl (150 kB)
Downloading pox-0.3.7-py3-none-any.whl (29 kB)
Downloading ppft-1.7.8-py3-none-any.whl (56 kB)
Using cached psutil-7.2.2-cp36-abi3-manylinux2010_x86_64.manylinux_2_12_x86_64.manylinux_2_28_x86_64.whl (155 kB)
Using cached schema-0.7.8-py2.py3-none-any.whl (19 kB)
Using cached tqdm-4.67.3-py3-none-any.whl (78 kB)
Using cached uvicorn-0.48.0-py3-none-any.whl (71 kB)
Using cached click-8.4.1-py3-none-any.whl (116 kB)

Using cached h11-0.16.0-py3-none-any.whl (37 kB)

Installing collected packages: schema, antlr4-python3-runtime, zipp, urllib3, typing-extensions, tqdm, tblib, smdebug-rulesconfig, six, rpds-py, pyyaml, pygments, psutil, protobuf, ppft, pox, platformdirs, packaging, numpy, mock, mdurl, jmespath, idna, h11, graphql-core, dill, cloudpickle, click, charset_normalizer, certifi, attrs, annotated-types, annotated-doc, uvicorn, typing-inspection, requests, referencing, python-dateutil, pydantic-core, omegaconf, multiprocessing, markdown-it-py, importlib-metadata, graphql-relay, google-pasta, anyio, starlette, rich, pydantic, pathos, pandas, jsonschema-specifications, graphene, docker, botocore, s3transfer, jsonschema, fastapi, boto3, sagemaker-core, sagemaker

Attempting uninstall: schema

Found existing installation: schema 0.7.8

Uninstalling schema-0.7.8:

Successfully uninstalled schema-0.7.8

Attempting uninstall: antlr4-python3-runtime

Found existing installation: antlr4-python3-runtime 4.9.3

Uninstalling antlr4-python3-runtime-4.9.3:

Successfully uninstalled antlr4-python3-runtime-4.9.3

Attempting uninstall: zipp

2/61 [zipp]

Found existing installation: zipp 4.1.0

2/61 [zipp]

Uninstalling zipp-4.1.0:

2/61 [zipp]

Successfully uninstalled zipp-4.1.0

2/61 [zipp]

Attempting uninstall: urllib3

2/61 [zipp]

Found existing installation: urllib3 2.7.0

2/61 [zipp]

Uninstalling urllib3-2.7.0:

2/61 [zipp]

Successfully uninstalled urllib3-2.7.0

2/61 [zipp]

Attempting uninstall: typing-extensions

2/61 [zipp]
Found existing installation: typing_extensions 4.15.0

2/61 [zipp]
Uninstalling typing_extensions-4.15.0:

2/61 [zipp]

4/61 [typing-extensions]

4/61 [typing-extensions]

4/61 [typing-extensions]

4/61 [typing-extensions]

4/61 [typing-extensions]

4/61 [typing-extensions]

Successfully uninstalled typing_extensions-4.15.0

4/61 [typing-extensions]

Attempting uninstall: tqdm

4/61 [typing-extensions]
Found existing installation: tqdm 4.67.3

4/61 [typing-extensions]

5/61 [tqdm]

Uninstalling tqdm-4.67.3:

5/61 [tqdm]

Successfully uninstalled tqdm-4.67.3

5/61 [tqdm]

Attempting uninstall: tblib

5/61 [tqdm]

Found existing installation: tblib 3.2.2

5/61 [tqdm]

Uninstalling tblib-3.2.2:

5/61 [tqdm]

Successfully uninstalled tblib-3.2.2

5/61 [tqdm]

Attempting uninstall: smdebug-rulesconfig

5/61 [tqdm]

Found existing installation: smdebug-rulesconfig 1.0.1

5/61 [tqdm]

Uninstalling smdebug-rulesconfig-1.0.1:

5/61 [tqdm]

Successfully uninstalled smdebug-rulesconfig-1.0.1

5/61 [tqdm]

Attempting uninstall: six

5/61 [tqdm]

8/61 [six]
Found existing installation: six 1.17.0

8/61 [six]
Uninstalling six-1.17.0:

8/61 [six]

8/61 [six]

8/61 [six]

8/61 [six]

8/61 [six]

8/61 [six]

Successfully uninstalled six-1.17.0

8/61 [six]
Attempting uninstall: rpds-py

8/61 [six]

9/61 [rpds-py]
Found existing installation: rpds-py 0.30.0

9/61 [rpds-py]
Uninstalling rpds-py-0.30.0:

```
9/61 [rpds-py]
    Successfully uninstalled rpds-py-0.30.0

9/61 [rpds-py]
    Attempting uninstall: pyyaml

9/61 [rpds-py]

    Found existing installation: PyYAML 6.0.3

9/61 [rpds-py]
    Uninstalling PyYAML-6.0.3:

9/61 [rpds-py]
    Successfully uninstalled PyYAML-6.0.3

9/61 [rpds-py]

    Attempting uninstall: pygments

9/61 [rpds-py]
    Found existing installation: Pygments 2.20.0

9/61 [rpds-py]

    Uninstalling Pygments-2.20.0:

9/61 [rpds-py]
    Successfully uninstalled Pygments-2.20.0

9/61 [rpds-py]

11/61 [pygments]

11/61 [pygments]
```

11/61 [pygments]

11/61 [pygments]

11/61 [pygments]

Attempting uninstall: psutil

11/61 [pygments]

Found existing installation: psutil 7.2.2

11/61 [pygments]

Uninstalling psutil-7.2.2:

11/61 [pygments]

Successfully uninstalled psutil-7.2.2

11/61 [pygments]

Attempting uninstall: protobuf

11/61 [pygments]

Found existing installation: protobuf 6.33.6

11/61 [pygments]

Uninstalling protobuf-6.33.6:

11/61 [pygments]

Successfully uninstalled protobuf-6.33.6

11/61 [pygments]

13/61 [protobuf]

14/61 [ppft]

Attempting uninstall: platformdirs

14/61 [ppft]

Found existing installation: platformdirs 4.9.6

14/61 [ppft]

Uninstalling platformdirs-4.9.6:

14/61 [ppft]

Successfully uninstalled platformdirs-4.9.6

14/61 [ppft]

Attempting uninstall: packaging

14/61 [ppft]

Found existing installation: packaging 26.2

14/61 [ppft]

Uninstalling packaging-26.2:

14/61 [ppft]

Successfully uninstalled packaging-26.2

14/61 [ppft]

Attempting uninstall: numpy

14/61 [ppft]

18/61 [numpy]

Found existing installation: numpy 2.4.6

18/61 [numpy]

Uninstalling numpy-2.4.6:

18/61 [numpy]

Successfully uninstalled numpy-2.4.6

18/61 [numpy]

18/61 [numpy]

18/61 [numpy]

18/61 [numpy]

18/61 [numpy]

18/61 [numpy]

18/61 [numpy]

18/61 [numpy]

18/61 [numpy]

18/61 [numpy]

18/61 [numpy]

18/61 [numpy]

18/61 [numpy]

Attempting uninstall: mock

18/61 [numpy]

Found existing installation: mock 4.0.3

18/61 [numpy]

Uninstalling mock-4.0.3:

18/61 [numpy]

Successfully uninstalled mock-4.0.3

18/61 [numpy]

```
Attempting uninstall: mdurl

18/61 [numpy]
    Found existing installation: mdurl 0.1.2

18/61 [numpy]
    Uninstalling mdurl-0.1.2:

18/61 [numpy]
    Successfully uninstalled mdurl-0.1.2

18/61 [numpy]
    Attempting uninstall: jmespath

18/61 [numpy]
    Found existing installation: jmespath 1.1.0

18/61 [numpy]

    Uninstalling jmespath-1.1.0:

18/61 [numpy]
    Successfully uninstalled jmespath-1.1.0

18/61 [numpy]

21/61 [jmespath]
    Attempting uninstall: idna

21/61 [jmespath]

    Found existing installation: idna 3.16

21/61 [jmespath]
    Uninstalling idna-3.16:

21/61 [jmespath]
    Successfully uninstalled idna-3.16

21/61 [jmespath]

    Attempting uninstall: h11

21/61 [jmespath]
    Found existing installation: h11 0.16.0
```

```
21/61 [jmespath]
    Uninstalling h11-0.16.0:

21/61 [jmespath]
    Successfully uninstalled h11-0.16.0

21/61 [jmespath]
    Attempting uninstall: graphql-core

21/61 [jmespath]

24/61 [graphql-core]

    Found existing installation: graphql-core 3.2.8

24/61 [graphql-core]
    Uninstalling graphql-core-3.2.8:

24/61 [graphql-core]
    Successfully uninstalled graphql-core-3.2.8

24/61 [graphql-core]

24/61 [graphql-core]

    Attempting uninstall: dill

24/61 [graphql-core]
    Found existing installation: dill 0.4.1

24/61 [graphql-core]
    Uninstalling dill-0.4.1:

24/61 [graphql-core]
    Successfully uninstalled dill-0.4.1

24/61 [graphql-core]

25/61 [dill]
    Attempting uninstall: cloudpickle

25/61 [dill]
    Found existing installation: cloudpickle 3.1.2
```



```
25/61 [dill]
    Uninstalling cloudpickle-3.1.2:

25/61 [dill]
    Successfully uninstalled cloudpickle-3.1.2

25/61 [dill]

    Attempting uninstall: click

25/61 [dill]
    Found existing installation: click 8.4.1

25/61 [dill]
    Uninstalling click-8.4.1:

25/61 [dill]
    Successfully uninstalled click-8.4.1

25/61 [dill]

    Attempting uninstall: charset_normalizer

25/61 [dill]
    Found existing installation: charset-normalizer 3.4.7

25/61 [dill]
    Uninstalling charset-normalizer-3.4.7:

25/61 [dill]

28/61 [charset_normalizer]

28/61 [charset_normalizer]

28/61 [charset_normalizer]

28/61 [charset_normalizer]

28/61 [charset_normalizer]
```

28/61 [charset_normalizer]

Successfully uninstalled charset-normalizer-3.4.7

28/61 [charset_normalizer]

Attempting uninstall: certifi

28/61 [charset_normalizer]

Found existing installation: certifi 2026.5.20

28/61 [charset_normalizer]

Uninstalling certifi-2026.5.20:

28/61 [charset_normalizer]

Successfully uninstalled certifi-2026.5.20

28/61 [charset_normalizer]

29/61 [certifi]

Attempting uninstall: attrs

29/61 [certifi]

Found existing installation: attrs 26.1.0

29/61 [certifi]

Uninstalling attrs-26.1.0:

29/61 [certifi]

Successfully uninstalled attrs-26.1.0

29/61 [certifi]

Attempting uninstall: annotated-types

29/61 [certifi]

Found existing installation: annotated-types 0.7.0

29/61 [certifi]

Uninstalling annotated-types-0.7.0:

29/61 [certifi]

Successfully uninstalled annotated-types-0.7.0

```
29/61 [certifi]
    Attempting uninstall: annotated-doc

29/61 [certifi]


32/61 [annotated-doc]
    Found existing installation: annotated-doc 0.0.4

32/61 [annotated-doc]
    Uninstalling annotated-doc-0.0.4:

32/61 [annotated-doc]
    Successfully uninstalled annotated-doc-0.0.4

32/61 [annotated-doc]
    Attempting uninstall: uvicorn

32/61 [annotated-doc]
    Found existing installation: uvicorn 0.48.0

32/61 [annotated-doc]

    Uninstalling uvicorn-0.48.0:

32/61 [annotated-doc]
    Successfully uninstalled uvicorn-0.48.0

32/61 [annotated-doc]

    Attempting uninstall: typing-inspection

32/61 [annotated-doc]
    Found existing installation: typing-inspection 0.4.2

32/61 [annotated-doc]
    Uninstalling typing-inspection-0.4.2:

32/61 [annotated-doc]

    Successfully uninstalled typing-inspection-0.4.2

32/61 [annotated-doc]
    Attempting uninstall: requests
```

```
32/61 [annotated-doc]

35/61 [requests]
    Found existing installation: requests 2.34.2

35/61 [requests]
    Uninstalling requests-2.34.2:

35/61 [requests]
    Successfully uninstalled requests-2.34.2

35/61 [requests]

    Attempting uninstall: referencing

35/61 [requests]

    Found existing installation: referencing 0.37.0

35/61 [requests]
    Uninstalling referencing-0.37.0:

35/61 [requests]
    Successfully uninstalled referencing-0.37.0

35/61 [requests]

    Attempting uninstall: python-dateutil

35/61 [requests]
    Found existing installation: python-dateutil 2.9.0.post0

35/61 [requests]
    Uninstalling python-dateutil-2.9.0.post0:

35/61 [requests]
    Successfully uninstalled python-dateutil-2.9.0.post0

35/61 [requests]

37/61 [python-dateutil]
    Attempting uninstall: pydantic-core

37/61 [python-dateutil]
```

```
Found existing installation: pydantic_core 2.46.4

37/61 [python-dateutil]
  Uninstalling pydantic_core-2.46.4:

37/61 [python-dateutil]
  Successfully uninstalled pydantic_core-2.46.4

37/61 [python-dateutil]

Attempting uninstall: omegaconf

37/61 [python-dateutil]
  Found existing installation: omegaconf 2.3.0

37/61 [python-dateutil]
  Uninstalling omegaconf-2.3.0:

37/61 [python-dateutil]
  Successfully uninstalled omegaconf-2.3.0

37/61 [python-dateutil]

39/61 [omegaconf]

Attempting uninstall: multiprocessing

39/61 [omegaconf]
  Found existing installation: multiprocessing 0.70.19

39/61 [omegaconf]
  Uninstalling multiprocessing-0.70.19:

39/61 [omegaconf]
  Successfully uninstalled multiprocessing-0.70.19

39/61 [omegaconf]

40/61 [multiprocessing]
  Attempting uninstall: markdown-it-py

40/61 [multiprocessing]
```

```
Found existing installation: markdown-it-py 4.2.0

40/61 [multiprocess]
  Uninstalling markdown-it-py-4.2.0:

40/61 [multiprocess]
  Successfully uninstalled markdown-it-py-4.2.0

40/61 [multiprocess]

Attempting uninstall: importlib-metadata

40/61 [multiprocess]

42/61 [importlib-metadata]
  Found existing installation: importlib-metadata 6.11.0

42/61 [importlib-metadata]
  Uninstalling importlib-metadata-6.11.0:

42/61 [importlib-metadata]
  Successfully uninstalled importlib-metadata-6.11.0

42/61 [importlib-metadata]

Attempting uninstall: graphql-relay

42/61 [importlib-metadata]
  Found existing installation: graphql-relay 3.2.0

42/61 [importlib-metadata]
  Uninstalling graphql-relay-3.2.0:

42/61 [importlib-metadata]
  Successfully uninstalled graphql-relay-3.2.0

42/61 [importlib-metadata]
  Attempting uninstall: google-pasta

42/61 [importlib-metadata]

Found existing installation: google-pasta 0.2.0

42/61 [importlib-metadata]
  Uninstalling google-pasta-0.2.0:
```

```
42/61 [importlib-metadata]  
    Successfully uninstalled google-pasta-0.2.0
```

```
42/61 [importlib-metadata]
```

```
    Attempting uninstall: anyio
```

```
42/61 [importlib-metadata]  
    Found existing installation: anyio 4.13.0
```

```
42/61 [importlib-metadata]  
    Uninstalling anyio-4.13.0:
```

```
42/61 [importlib-metadata]  
    Successfully uninstalled anyio-4.13.0
```

```
42/61 [importlib-metadata]
```

```
45/61 [anyio]
```

```
    Attempting uninstall: starlette
```

```
45/61 [anyio]  
    Found existing installation: starlette 0.52.1
```

```
45/61 [anyio]  
    Uninstalling starlette-0.52.1:
```

```
45/61 [anyio]  
    Successfully uninstalled starlette-0.52.1
```

```
45/61 [anyio]
```

```
46/61 [starlette]
```

```
    Attempting uninstall: rich
```

```
46/61 [starlette]  
    Found existing installation: rich 14.3.4
```

```
46/61 [starlette]  
    Uninstalling rich-14.3.4:
```

```
46/61 [starlette]
```

Successfully uninstalled rich-14.3.4

46/61 [starlette]

47/61 [rich]

Attempting uninstall: pydantic

47/61 [rich]

48/61 [pydantic]

Found existing installation: pydantic 2.13.4

48/61 [pydantic]

Uninstalling pydantic-2.13.4:

48/61 [pydantic]

Successfully uninstalled pydantic-2.13.4

48/61 [pydantic]

48/61 [pydantic]

49/61 [pathos]

Attempting uninstall: pandas

49/61 [pathos]

Found existing installation: pandas 2.3.3

49/61 [pathos]

50/61 [pandas]

Uninstalling pandas-2.3.3:

50/61 [pandas]

Successfully uninstalled pandas-2.3.3

50/61 [pandas]

50/61 [pandas]

50/61 [pandas]

50/61 [pandas]

50/61 [pandas]

50/61 [pandas]

50/61 [pandas]

50/61 [pandas]

50/61 [pandas]

50/61 [pandas]

50/61 [pandas]

50/61 [pandas]

50/61 [pandas]

50/61 [pandas]

50/61 [pandas]

50/61 [pandas]

50/61 [pandas]

50/61 [pandas]

50/61 [pandas]

50/61 [pandas]

50/61 [pandas]

50/61 [pandas]

50/61 [pandas]

50/61 [pandas]

50/61 [pandas]

50/61 [pandas]

50/61 [pandas]

50/61 [pandas]

50/61 [pandas]

Attempting uninstall: jsonschema-specifications

50/61 [pandas]

Found existing installation: jsonschema-specifications 2025.9.1

50/61 [pandas]

Uninstalling jsonschema-specifications-2025.9.1:

50/61 [pandas]

Successfully uninstalled jsonschema-specifications-2025.9.1

50/61 [pandas]

Attempting uninstall: graphene

50/61 [pandas]

52/61 [graphene]

Found existing installation: graphene 3.4.3

52/61 [graphene]

Uninstalling graphene-3.4.3:

52/61 [graphene]

Successfully uninstalled graphene-3.4.3

52/61 [graphene]

Attempting uninstall: docker

52/61 [graphene]

53/61 [docker]

Found existing installation: docker 7.1.0

53/61 [docker]

Uninstalling docker-7.1.0:

```
53/61 [docker]
    Successfully uninstalled docker-7.1.0

53/61 [docker]

    Attempting uninstall: botocore

53/61 [docker]
    Found existing installation: botocore 1.43.14

53/61 [docker]


54/61 [botocore]

    Uninstalling botocore-1.43.14:

54/61 [botocore]
    Successfully uninstalled botocore-1.43.14

54/61 [botocore]


54/61 [botocore]


54/61 [botocore]


54/61 [botocore]


54/61 [botocore]


54/61 [botocore]


    Attempting uninstall: s3transfer

54/61 [botocore]
    Found existing installation: s3transfer 0.17.0

54/61 [botocore]
    Uninstalling s3transfer-0.17.0:

54/61 [botocore]
```

Successfully uninstalled s3transfer-0.17.0

54/61 [botocore]

55/61 [s3transfer]

Attempting uninstall: jsonschema

55/61 [s3transfer]

Found existing installation: jsonschema 4.26.0

55/61 [s3transfer]

Uninstalling jsonschema-4.26.0:

55/61 [s3transfer]

Successfully uninstalled jsonschema-4.26.0

55/61 [s3transfer]

Attempting uninstall: fastapi

55/61 [s3transfer]

57/61 [fastapi]

Found existing installation: fastapi 0.136.3

57/61 [fastapi]

Uninstalling fastapi-0.136.3:

57/61 [fastapi]

Successfully uninstalled fastapi-0.136.3

57/61 [fastapi]

Attempting uninstall: boto3

57/61 [fastapi]

Found existing installation: boto3 1.43.14

57/61 [fastapi]

Uninstalling boto3-1.43.14:

57/61 [fastapi]

Successfully uninstalled boto3-1.43.14

57/61 [fastapi]

58/61 [boto3]

Attempting uninstall: sagemaker-core

58/61 [boto3]

Found existing installation: sagemaker-core 2.12.0

58/61 [boto3]

Uninstalling sagemaker-core-2.12.0:

58/61 [boto3]

Successfully uninstalled sagemaker-core-2.12.0

58/61 [boto3]

59/61 [sagemaker-core]

59/61 [sagemaker-core]

Attempting uninstall: sagemaker

59/61 [sagemaker-core]

Found existing installation: sagemaker 3.12.0

59/61 [sagemaker-core]

Uninstalling sagemaker-3.12.0:

59/61 [sagemaker-core]

Successfully uninstalled sagemaker-3.12.0

59/61 [sagemaker-core]

60/61

[sagemaker]

60/61

[sagemaker]

60/61

[sagemaker]

60/61

[sagemaker]

60/61

[sagemaker]

60/61

[sagemaker]

61/61

[sagemaker]

ERROR: pip's dependency resolver does not currently take into account all the packages that are installed. This behaviour is the source of the following dependency conflicts.

autogluon-multimodal 1.5.0 requires nvidia-ml-py3<8.0,>=7.352.0, which is not installed.

autogluon-timeseries 1.5.0 requires chronos-forecasting<2.4,>=2.2.2, which is not installed.

autogluon-timeseries 1.5.0 requires einops<1,>=0.7, which is not installed.

autogluon-timeseries 1.5.0 requires peft<0.18,>=0.13.0, which is not installed.

dash 2.18.1 requires dash-core-components==2.0.0, which is not installed.

dash 2.18.1 requires dash-html-components==2.0.0, which is not installed.

dash 2.18.1 requires dash-table==5.0.0, which is not installed.

aiobotocore 3.6.0 requires botocore<1.43.1,>=1.42.90, but you have botocore 1.43.14 which is incompatible.

amazon-sagemaker-sql-editor 0.2.4 requires rpds-py==0.27.1, but you have rpds-py 0.30.0 which is incompatible.

amazon-sagemaker-sql-magic 0.1.4 requires numpy<2, but you have numpy 2.4.6 which is incompatible.

amazon-sagemaker-sql-magic 0.1.4 requires sqlparse==0.5.0, but you have sqlparse 0.5.5 which is incompatible.

autogluon-common 1.5.0 requires numpy<2.4.0,>=1.25.0, but you have numpy 2.4.6 which is incompatible.

autogluon-common 1.5.0 requires pandas<2.4.0,>=2.0.0, but you have pandas 3.0.3 which is incompatible.

autogluon-common 1.5.0 requires psutil<7.2.0,>=5.7.3, but you have psutil 7.2.2 which is incompatible.

autogluon-common 1.5.0 requires pyarrow<21.0.0,>=7.0.0, but you have pyarrow 23.0.1 which is incompatible.

autogluon-core 1.5.0 requires numpy<2.4.0,>=1.25.0, but you have numpy 2.4.6 which is incompatible.

autogluon-core 1.5.0 requires pandas<2.4.0,>=2.0.0, but you have pandas 3.0.3 which is incompatible.

autogluon-core 1.5.0 requires scikit-learn<1.8.0,>=1.4.0, but you have scikit-learn 1.8.0 which is incompatible.

autogluon-core 1.5.0 requires scipy<1.17,>=1.5.4, but you have scipy 1.17.1 which is incompatible.

Successfully installed annotated-doc-0.0.4 annotated-types-0.7.0
 antlr4-python3-runtime-4.9.3 anyio-4.13.0 attrs-25.4.0 boto3-1.43.14
 botocore-1.43.14 certifi-2026.5.20 charset_normalizer-3.4.7 click-8.4.1
 cloudpickle-3.1.2 dill-0.4.1 docker-7.1.0 fastapi-0.136.3 google-pasta-0.2.0
 graphene-3.4.3 graphql-core-3.2.8 graphql-relay-3.2.0 h11-0.16.0 idna-3.16
 importlib-metadata-6.11.0 jmespath-1.1.0 jsonschema-4.26.0 jsonschema-
 specifications-2025.9.1 markdown-it-py-4.2.0 mdurl-0.1.2 mock-4.0.3
 multiprocessing-0.70.19 numpy-2.4.6 omegaconf-2.3.0 packaging-24.2 pandas-3.0.3
 pathos-0.3.5 platformdirs-4.9.6 pox-0.3.7 ppft-1.7.8 protobuf-6.33.6
 psutil-7.2.2 pydantic-2.13.4 pydantic-core-2.46.4 pygments-2.20.0 python-
 dateutil-2.9.0.post0 pyyaml-6.0.3 referencing-0.37.0 requests-2.34.2 rich-14.3.4
 rpds-py-0.30.0 s3transfer-0.17.0 sagemaker-2.257.3 sagemaker-core-1.0.78
 schema-0.7.8 six-1.17.0 smdebug-rulesconfig-1.0.1 starlette-1.1.0 tblib-3.2.2
 tqdm-4.67.3 typing-extensions-4.15.0 typing-inspection-0.4.2 urllib3-2.7.0
 uvicorn-0.48.0 zipp-4.1.0

Note: you may need to restart the kernel to use updated packages.

```
[2]: import boto3
import sagemaker
from sagemaker.session import Session
from sagemaker import get_execution_role
import pandas as pd
import numpy as np
import time
from time import gmtime, strftime, sleep
from sagemaker.feature_store.feature_group import FeatureGroup
import pprint
```

sagemaker.config INFO - Not applying SDK defaults from location:
 /etc/xdg/sagemaker/config.yaml

sagemaker.config INFO - Not applying SDK defaults from location:
 /home/sagemaker-user/.config/sagemaker/config.yaml

```
[3]: print("sagemaker version:", sagemaker.__version__) # expect something like 2.
      ↪254.x
```

sagemaker version: 2.257.3

```
[4]: original_boto3_version = boto3.__version__
      ↪%pip install -q 'boto3>1.17.21'
```

```
[5]: region = boto3.Session().region_name
boto_session = boto3.Session(region_name=region)

sagemaker_client = boto_session.client("sagemaker", region_name=region)
featurestore_runtime = boto_session.client("sagemaker-featurestore-runtime",
      ↪region_name=region)
```

```

feature_store_session = Session(
    boto_session=boto_session,
    sagemaker_client=sagemaker_client,
    sagemaker_featurestore_runtime_client=featurestore_runtime,
)

default_s3_bucket_name = feature_store_session.default_bucket()
prefix = "sagemaker-featurestore-homework-3-1"
print("Region :", region)
print("Bucket :", default_s3_bucket_name)

```

Region : us-east-1
 Bucket : sagemaker-us-east-1-697214847177

```

[6]: role = get_execution_role()
print("Role :", role)

```

Role : arn:aws:iam::697214847177:role/LabRole

```

[7]: HOUSING_PATH = "housing.csv"
GMAPS_PATH = "housing_gmaps_data_raw.csv"

housing = pd.read_csv(HOUSING_PATH)
gmaps = pd.read_csv(GMAPS_PATH)

print("housing:", housing.shape, " gmaps:", gmaps.shape)
housing.head()

```

housing: (20640, 10) gmaps: (12590, 30)

```

[7]:
  longitude  latitude  housing_median_age  total_rooms  total_bedrooms  \
0   -122.23    37.88                41.0         880.0          129.0
1   -122.22    37.86                21.0        7099.0         1106.0
2   -122.24    37.85                52.0        1467.0          190.0
3   -122.25    37.85                52.0        1274.0          235.0
4   -122.25    37.85                52.0        1627.0          280.0

  population  households  median_income  median_house_value  ocean_proximity
0       322.0       126.0         8.3252         452600.0         NEAR BAY
1      2401.0      1138.0         8.3014        358500.0         NEAR BAY
2       496.0       177.0         7.2574        352100.0         NEAR BAY
3       558.0       219.0         5.6431        341300.0         NEAR BAY
4       565.0       259.0         3.8462        342200.0         NEAR BAY

```

```

[8]: gmaps_slim = (gmaps[["latitude", "longitude", "neighborhood-political",
↪ "postal_code"]]
        .drop_duplicates(subset=["latitude", "longitude"]))

```

```

df = housing.merge(gmaps_slim, on=["latitude", "longitude"], how="left")

# neighborhood label
df = df[df["neighborhood-political"].notna()].copy()

print("Rows after join + neighborhood filter:", len(df))
print("Distinct neighborhoods          :", df['neighborhood-political'].
      ↪nunique())

```

Rows after join + neighborhood filter: 9000
 Distinct neighborhoods : 1306

```

[9]: # one-hot encode ocean_proximity with all 5 categories enforced
ALL_OCEAN = ["<1H OCEAN", "INLAND", "ISLAND", "NEAR BAY", "NEAR OCEAN"]
df["ocean_proximity"] = pd.Categorical(df["ocean_proximity"],
    ↪categories=ALL_OCEAN)

ocean_oh = pd.get_dummies(df["ocean_proximity"]).astype(int)
ocean_oh.columns = (ocean_oh.columns.str.lower()
    ↪.str.replace("<1h ocean", "lt_1h_ocean",
    ↪regex=False)
    ↪.str.replace(" ", "_"))
df = pd.concat([df, ocean_oh], axis=1)

OH_COLS = list(ocean_oh.columns)
print("One-hot columns:", OH_COLS)

```

One-hot columns: ['lt_1h_ocean', 'inland', 'island', 'near_bay', 'near_ocean']

```

[10]: # bedrooms_per_household at the row level
df["bedrooms_per_household"] = df["total_bedrooms"] / df["households"]

# Impute: where the ratio is NaN (because total_bedrooms was null),
# fill with the mean ratio for that row's postal code
df["bedrooms_per_household"] = (df.
    ↪groupby("postal_code")["bedrooms_per_household"]
    ↪.transform(lambda s: s.fillna(s.mean()))

# Safety net: a postal code with zero non-null ratios would leave residual NaNs
df["bedrooms_per_household"] = df["bedrooms_per_household"].fillna(
    df["bedrooms_per_household"].mean()
)

print("Residual nulls in bedrooms_per_household:", df["bedrooms_per_household"].
    ↪isna().sum())

```

Residual nulls in bedrooms_per_household: 0

```
[11]: agg = (df.groupby("neighborhood-political")
            .agg(**{c: (c, "max") for c in OH_COLS},
                 median_house_value = ("median_house_value", "mean"),
                 median_house_age_raw = ("housing_median_age", "mean"),
                 total_households = ("households", "mean"),
                 bedrooms_per_household = ("bedrooms_per_household", "mean"))
            .reset_index()
            .rename(columns={"neighborhood-political": "neighborhood"}))

# Cap median_house_value at 500,000
agg["median_house_value"] = agg["median_house_value"].clip(upper=500000)

# Round households up to integer
agg["total_households"] = np.ceil(agg["total_households"]).astype(int)

# Bucket the averaged median house age into 10year ranges
def age_bucket(avg_age):
    lo = int(avg_age // 10) * 10
    return f"{lo}-{lo+9}"

agg["median_house_age"] = agg["median_house_age_raw"].apply(age_bucket)
agg = agg.drop(columns=["median_house_age_raw"])

# Final column order
neighborhood_features = agg[
    ["neighborhood"] + OH_COLS +
    ["median_house_value", "median_house_age", "total_households",
     ↪ "bedrooms_per_household"]
].copy()

print("Feature group shape:", neighborhood_features.shape)
print("Any nulls?", neighborhood_features.isna().any().any())
neighborhood_features.head()
```

Feature group shape: (1306, 10)

Any nulls? False

```
[11]:
```

	neighborhood	lt_1h_ocean	inland	island	near_bay	\
0	28 Palms	1	0	0	0	
1	Acorn Industrial	0	0	0	1	
2	Adams Hill	1	0	0	0	
3	Agua Mansa Industrial Corridor	0	1	0	0	
4	Al Tahoe	0	1	0	0	

	near_ocean	median_house_value	median_house_age	total_households	\
0	0	222200.000000	20-29	923	
1	0	81300.000000	50-59	147	

2	0	250733.333333	30-39	494
3	0	112300.000000	10-19	516
4	0	109180.000000	20-29	249

	bedrooms_per_household
0	1.017335
1	1.659864
2	1.034649
3	1.102713
4	1.641739

```
[12]: #Building the Feature Group
```

```
[13]: def cast_object_to_string(frame):
    for col in frame.columns:
        if frame.dtypes[col] == "object":
            frame[col] = frame[col].astype("str").astype("string")

cast_object_to_string(neighborhood_features)

record_identifier_feature_name = "neighborhood"
event_time_feature_name       = "event_time"

current_time_sec = float(round(time.time()))
neighborhood_features[event_time_feature_name] = pd.Series(
    [current_time_sec] * len(neighborhood_features), dtype="float64"
)

neighborhood_features.dtypes
```

```
[13]: neighborhood          str
lt_1h_ocean                int64
inland                     int64
island                     int64
near_bay                   int64
near_ocean                 int64
median_house_value         float64
median_house_age           str
total_households           int64
bedrooms_per_household     float64
event_time                 float64
dtype: object
```

```
[16]: for col in ["neighborhood", "median_house_age"]:
    neighborhood_features[col] = neighborhood_features[col].astype(object)

print(neighborhood_features.dtypes) # verify
```

```

neighborhood          object
lt_1h_ocean           int64
inland                int64
island                int64
near_bay              int64
near_ocean            int64
median_house_value    float64
median_house_age      object
total_households      int64
bedrooms_per_household float64
event_time            float64
dtype: object

```

```

[17]: neighborhood_feature_group_name = "neighborhood-feature-group-" + \
    ↳strftime("%d-%H-%M-%S", gmtime())
    print("FeatureGroup name:", neighborhood_feature_group_name)

    neighborhood_feature_group = FeatureGroup(
        name=neighborhood_feature_group_name,
        sagemaker_session=feature_store_session,
    )

    neighborhood_feature_group.
    ↳load_feature_definitions(data_frame=neighborhood_features)

```

FeatureGroup name: neighborhood-feature-group-25-14-30-23

```

[17]: [FeatureDefinition(feature_name='neighborhood',
    feature_type=<FeatureTypeEnum.STRING: 'String'>, collection_type=None),
    FeatureDefinition(feature_name='lt_1h_ocean',
    feature_type=<FeatureTypeEnum.INTEGRAL: 'Integral'>, collection_type=None),
    FeatureDefinition(feature_name='inland',
    feature_type=<FeatureTypeEnum.INTEGRAL: 'Integral'>, collection_type=None),
    FeatureDefinition(feature_name='island',
    feature_type=<FeatureTypeEnum.INTEGRAL: 'Integral'>, collection_type=None),
    FeatureDefinition(feature_name='near_bay',
    feature_type=<FeatureTypeEnum.INTEGRAL: 'Integral'>, collection_type=None),
    FeatureDefinition(feature_name='near_ocean',
    feature_type=<FeatureTypeEnum.INTEGRAL: 'Integral'>, collection_type=None),
    FeatureDefinition(feature_name='median_house_value',
    feature_type=<FeatureTypeEnum.FRACTIONAL: 'Fractional'>, collection_type=None),
    FeatureDefinition(feature_name='median_house_age',
    feature_type=<FeatureTypeEnum.STRING: 'String'>, collection_type=None),
    FeatureDefinition(feature_name='total_households',
    feature_type=<FeatureTypeEnum.INTEGRAL: 'Integral'>, collection_type=None),
    FeatureDefinition(feature_name='bedrooms_per_household',
    feature_type=<FeatureTypeEnum.FRACTIONAL: 'Fractional'>, collection_type=None),
    FeatureDefinition(feature_name='event_time',

```

```
feature_type=<FeatureTypeEnum.FRACTIONAL: 'Fractional'>, collection_type=None)]
```

```
[18]: def wait_for_feature_group_creation_complete(fg):
        status = fg.describe().get("FeatureGroupStatus")
        while status == "Creating":
            print("Waiting for FeatureGroup creation...")
            time.sleep(5)
            status = fg.describe().get("FeatureGroupStatus")
        if status != "Created":
            raise RuntimeError(f"Failed to create FeatureGroup {fg.name}")
        print(f"FeatureGroup {fg.name} successfully created.")

        neighborhood_feature_group.create(
            s3_uri=f"s3://{default_s3_bucket_name}/{prefix}",
            record_identifier_name=record_identifier_feature_name,
            event_time_feature_name=event_time_feature_name,
            role_arn=role,
            enable_online_store=True,
        )

        wait_for_feature_group_creation_complete(neighborhood_feature_group)
```

Waiting for FeatureGroup creation...

Waiting for FeatureGroup creation...

Waiting for FeatureGroup creation...

Waiting for FeatureGroup creation...

FeatureGroup neighborhood-feature-group-25-14-30-23 successfully created.

```
[19]: neighborhood_feature_group.describe()
```

```
[19]: {'FeatureGroupArn': 'arn:aws:sagemaker:us-east-1:697214847177:feature-
group/neighborhood-feature-group-25-14-30-23',
      'FeatureGroupName': 'neighborhood-feature-group-25-14-30-23',
      'RecordIdentifierFeatureName': 'neighborhood',
      'EventTimeFeatureName': 'event_time',
      'FeatureDefinitions': [{'FeatureName': 'neighborhood',
                              'FeatureType': 'String'},
                             {'FeatureName': 'lt_1h_ocean', 'FeatureType': 'Integral'},
                             {'FeatureName': 'inland', 'FeatureType': 'Integral'},
                             {'FeatureName': 'island', 'FeatureType': 'Integral'},
                             {'FeatureName': 'near_bay', 'FeatureType': 'Integral'},
                             {'FeatureName': 'near_ocean', 'FeatureType': 'Integral'},
                             {'FeatureName': 'median_house_value', 'FeatureType': 'Fractional'},
                             {'FeatureName': 'median_house_age', 'FeatureType': 'String'},
                             {'FeatureName': 'total_households', 'FeatureType': 'Integral'},
                             {'FeatureName': 'bedrooms_per_household', 'FeatureType': 'Fractional'}],
```

```

    {'FeatureName': 'event_time', 'FeatureType': 'Fractional'}],
    'CreationTime': datetime.datetime(2026, 5, 25, 14, 30, 27, 789000,
tzinfo=tzlocal()),
    'OnlineStoreConfig': {'EnableOnlineStore': True},
    'OfflineStoreConfig': {'S3StorageConfig': {'S3Uri': 's3://sagemaker-us-
east-1-697214847177/sagemaker-featurestore-homework-3-1',
    'ResolvedOutputS3Uri': 's3://sagemaker-us-east-1-697214847177/sagemaker-
featurestore-homework-3-1/697214847177/sagemaker/us-east-1/offline-
store/neighborhood-feature-group-25-14-30-23-1779719427/data'},
    'DisableGlueTableCreation': False,
    'DataCatalogConfig': {'TableName':
'neighborhood_feature_group_25_14_30_23_1779719427',
    'Catalog': 'AwsDataCatalog',
    'Database': 'sagemaker_featurestore'}}},
    'ThroughputConfig': {'ThroughputMode': 'OnDemand'},
    'RoleArn': 'arn:aws:iam::697214847177:role/LabRole',
    'FeatureGroupStatus': 'Created',
    'OnlineStoreTotalSizeBytes': 0,
    'ResponseMetadata': {'RequestId': 'f902bd14-5442-4579-bf02-7f604324dcc8',
    'HTTPStatusCode': 200,
    'HTTPHeaders': {'x-amzn-requestid': 'f902bd14-5442-4579-bf02-7f604324dcc8',
    'strict-transport-security': 'max-age=47304000; includeSubDomains',
    'x-frame-options': 'DENY',
    'content-security-policy': "frame-ancestors 'none'",
    'cache-control': 'no-cache, no-store, must-revalidate',
    'x-content-type-options': 'nosniff',
    'content-type': 'application/x-amz-json-1.1',
    'content-length': '2136',
    'date': 'Mon, 25 May 2026 14:31:48 GMT'},
    'RetryAttempts': 0}}

```

```

[20]: neighborhood_feature_group.ingest(
    data_frame=neighborhood_features,
    max_workers=3,
    wait=True,
)

```

```

[20]: IngestionManagerPandas(feature_group_name='neighborhood-feature-
group-25-14-30-23', feature_definitions={'neighborhood': {'FeatureName':
'neighborhood', 'FeatureType': 'String'}, 'lt_1h_ocean': {'FeatureName':
'lt_1h_ocean', 'FeatureType': 'Integral'}, 'inland': {'FeatureName': 'inland',
'FeatureType': 'Integral'}, 'island': {'FeatureName': 'island', 'FeatureType':
'Integral'}, 'near_bay': {'FeatureName': 'near_bay', 'FeatureType': 'Integral'},
'near_ocean': {'FeatureName': 'near_ocean', 'FeatureType': 'Integral'},
'median_house_value': {'FeatureName': 'median_house_value', 'FeatureType':
'Fractional'}, 'median_house_age': {'FeatureName': 'median_house_age',
'FeatureType': 'String'}, 'total_households': {'FeatureName':

```



```
'total_households', 'FeatureType': 'Integral'}, 'bedrooms_per_household':
{'FeatureName': 'bedrooms_per_household', 'FeatureType': 'Fractional'},
'event_time': {'FeatureName': 'event_time', 'FeatureType': 'Fractional'}},
sagemaker_fs_runtime_client_config=<botocore.config.Config object at
0x7f7b7bdf73e0>, sagemaker_session=<sagemaker.session.Session object at
0x7f7b7cea0110>, max_workers=3, max_processes=1, profile_name=None,
_async_result=<multiprocessing.pool.MapResult object at 0x7f7b79656420>,
_processing_pool=<pool ProcessPool(ncpus=1)>, _failed_indices=[])
```

```
[21]: def fetch(neighborhood_name):
    resp = featurestore_runtime.get_record(
        FeatureGroupName=neighborhood_feature_group_name,
        RecordIdentifierValueAsString=neighborhood_name,
    )
    print(f"=== {neighborhood_name} ===")
    pprint.pp(resp["Record"])
    print()
    return resp

_ = fetch("Brooktree")
```

```
=== Brooktree ===
[{'FeatureName': 'neighborhood', 'ValueAsString': 'Brooktree'},
 {'FeatureName': 'lt_1h_ocean', 'ValueAsString': '1'},
 {'FeatureName': 'inland', 'ValueAsString': '0'},
 {'FeatureName': 'island', 'ValueAsString': '0'},
 {'FeatureName': 'near_bay', 'ValueAsString': '0'},
 {'FeatureName': 'near_ocean', 'ValueAsString': '0'},
 {'FeatureName': 'median_house_value', 'ValueAsString': '257400.0'},
 {'FeatureName': 'median_house_age', 'ValueAsString': '0-9'},
 {'FeatureName': 'total_households', 'ValueAsString': '1438'},
 {'FeatureName': 'bedrooms_per_household',
  'ValueAsString': '1.0747777761942723'},
 {'FeatureName': 'event_time', 'ValueAsString': '1779718688.0'}]
```

```
[22]: _ = fetch("Fisherman's Wharf")
```

```
=== Fisherman's Wharf ===
[{'FeatureName': 'neighborhood', 'ValueAsString': "Fisherman's Wharf"},
 {'FeatureName': 'lt_1h_ocean', 'ValueAsString': '0'},
 {'FeatureName': 'inland', 'ValueAsString': '0'},
 {'FeatureName': 'island', 'ValueAsString': '0'},
 {'FeatureName': 'near_bay', 'ValueAsString': '1'},
 {'FeatureName': 'near_ocean', 'ValueAsString': '0'},
 {'FeatureName': 'median_house_value', 'ValueAsString': '500000.0'},
 {'FeatureName': 'median_house_age', 'ValueAsString': '50-59'},
 {'FeatureName': 'total_households', 'ValueAsString': '250'},
```

```
{'FeatureName': 'bedrooms_per_household', 'ValueAsString': '1.268'},
{'FeatureName': 'event_time', 'ValueAsString': '1779718688.0']}
```

```
[23]: _ = fetch("Los Osos")
```

```
=== Los Osos ===
[{'FeatureName': 'neighborhood', 'ValueAsString': 'Los Osos'},
 {'FeatureName': 'lt_1h_ocean', 'ValueAsString': '0'},
 {'FeatureName': 'inland', 'ValueAsString': '0'},
 {'FeatureName': 'island', 'ValueAsString': '0'},
 {'FeatureName': 'near_bay', 'ValueAsString': '0'},
 {'FeatureName': 'near_ocean', 'ValueAsString': '1'},
 {'FeatureName': 'median_house_value', 'ValueAsString': '221612.5'},
 {'FeatureName': 'median_house_age', 'ValueAsString': '10-19'},
 {'FeatureName': 'total_households', 'ValueAsString': '612'},
 {'FeatureName': 'bedrooms_per_household',
  'ValueAsString': '1.0478845404823531'},
 {'FeatureName': 'event_time', 'ValueAsString': '1779718688.0'}]
```

```
[24]: batch = featurestore_runtime.batch_get_record(Identifiers=[{
    "FeatureGroupName": neighborhood_feature_group_name,
    "RecordIdentifiersValueAsString": ["Brooktree", "Fisherman's Wharf", "Los_
↪Osos"],
}])

rows = []
for rec in batch["Records"]:
    rows.append({f["FeatureName"]: f["ValueAsString"] for f in rec["Record"]})

pd.DataFrame(rows).set_index("neighborhood")
```

```
[24]:
```

	lt_1h_ocean	inland	island	near_bay	near_ocean	\
neighborhood						
Brooktree	1	0	0	0	0	
Fisherman's Wharf	0	0	0	1	0	
Los Osos	0	0	0	0	1	

	median_house_value	median_house_age	total_households	\
neighborhood				
Brooktree	257400.0	0-9	1438	
Fisherman's Wharf	500000.0	50-59	250	
Los Osos	221612.5	10-19	612	

	bedrooms_per_household	event_time
neighborhood		
Brooktree	1.0747777761942723	1779718688.0

Fisherman's Wharf	1.268	1779718688.0
Los Osos	1.0478845404823531	1779718688.0

[]: